

CS07340 DESIGN AND ANALYSIS OF ALGORITHMS

WEEK 2 QUIZ (8 POINTS)

Name: _____

1. (4 points) The following are True/False questions about asymptotic notation. Clearly mark either T (for True) or F (for False).

- Big O notation creates an upper bound, that is, $O(n \log(n))$ means that the actual runtime/number of steps is in worst case bounded above by (constant times $n \log(n)$).
- Big O/ Θ / Ω notation is used up to scalar constants for the dominant term, that is, $\Theta(n^2)$ means “some quadratic such as $5n^2 + 5n - 3$ ”.
- Asymptotic calculations use calculus methods to arrive at an answer.
- If an algorithm solves a problem, then the runtime of the algorithm is the least amount of time needed to solve the problem. (That is, no other algorithm can be faster.)

2. (4 points) For each of the following functions, indicate how much the function’s value will change (i.e. doubles) if its argument n is increased fourfold.

- a) n^3
- b) $\log_2(n)$
- c) $1/n$
- d) n